Monthly Intelligence Report

Edition 002 — The NSW–VIC Interconnector Corridor

Where demand surges are outrunning transmission capacity. edition · May 2026 · SSI v4.0.2

CONTENTS

SECTION A

Executive Summary

SUBSTATIONS SCORED

8,500

Major + distribution across NEM & WA/NT

GRID LINES MAPPED

~9,200

~67,000 km transmission & distribution

MC SIMULATIONS

85.0 M

10,000 per substation

PUBLIC DATA SOURCES

20+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions. Its engine processes 146,000+ source data points per run, executes 85.0 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 1.78 billion arithmetic operations — producing 469,404 output data points with full uncertainty quantification across 8,500 substations and ~9,200 grid lines spanning ~67,000 km.

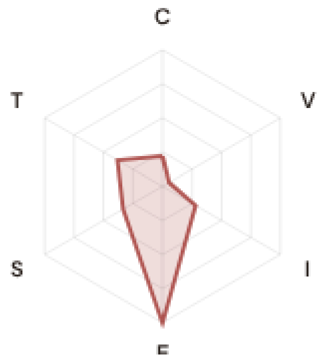
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 20+ verified public sources (AEMO, AER, ABS, BOM, CSIRO, Geoscience Australia, CER), making the methodology fully auditable and reproducible. Initially covering Australia's NEM and isolated systems, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Orange BSP 282

New South Wales · New South Wales · 33 kV

0.232	Low	0.271	1000th
R_FINAL	CLASSIFICATION	CI WIDTH	FLEET PERCENTILE



COMPONENTS

- **C** Continuity: 0.22 / 0.30 (22%)
- **I** Infrastructure: 0.28 / 0.25 (28%)
- **S** Saturation: 0.33 / 0.20 (33%)
- **V** Voltage: 0.06 / 0.10 (6%)
- **E** Economic: 1.47 / 0.10 (147%) ⚠
- **T** Transition: 0.38 / 0.05 (38%)

MODIFIERS

- **R3 Consequence:** 0.836 ↓ dampens
- **R4 Graph Criticality:** 0.871 ↓ dampens
- **R6a Restoration:** 0.906 ↓ dampens
- **R6b Seismic/Hazard:** 1.008 → neutral
- **R7 Digital Readiness:** 0.931 ↓ dampens

This month's spotlight — automatically selected for May 2026 — is **Orange BSP 282** in New South Wales, New South Wales. With an R_final of 0.232 (Low band, 1000th fleet percentile), its risk profile is dominated by the E (Economic) component at 1473% of its weight ceiling, followed by T (Transition) at 762%. Modifiers collectively shift R_base by -38.1%, reflecting the substation's specific geographic, socio-economic, and network context. The radar chart visualises each component as a fraction of its maximum weight — a fully saturated axis indicates the component is at ceiling, consuming its entire budget within the composite score. This substation's profile illustrates the SSI's core principle: risk is never mono-dimensional. Even when one component dominates, the interaction of all six — modulated by the five multipliers — determines the final score that drives investment prioritisation.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Australia's Energy Security Board and network operator strategic planning requires.

CYBER HIGH EXPOSURE	BLIND SPOTS	AVG R7 MODIFIER
0	711	1.0101
0.0% of fleet	High/Critical + R7 < median	Range [0.99, 1.05]

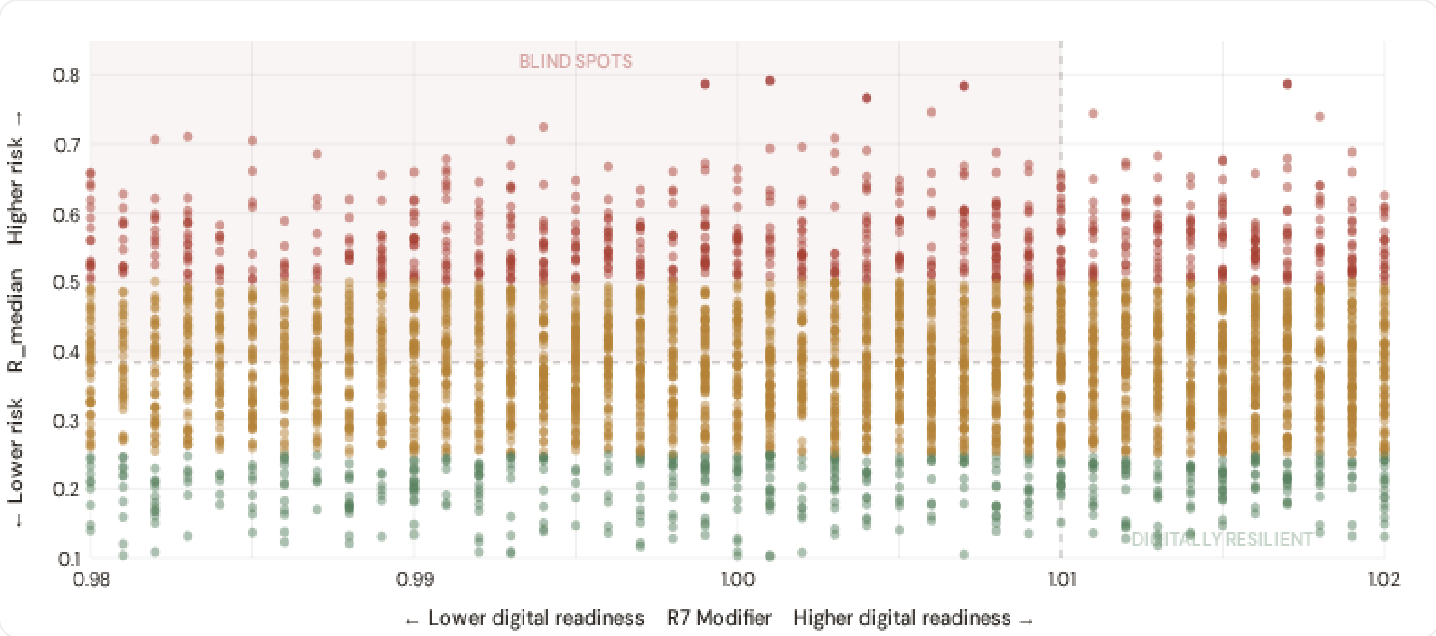
HIGH-CONSEQUENCE SUBS

21

0.2% · R3 ≥ 1.12

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **711 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 1.0100). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **New South Wales (211), Queensland (173), Victoria (111)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier — indicating robust digital infrastructure coverage concentrated in metropolitan networks (Ausgrid, CitiPower, Jemena).

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

🏭 Capital-Intensive / High-Consequence

Est. VoLL: Est. high VoLL

Heavy industry, critical infrastructure -- highest economic loss per interruption

2,146 substations (25.2%) High/Critical: **365** (17.0%) Avg R: **0.384** Avg E: **0.763** / 0.10

Low 277 Med 1504 High 361 Crit 4

🔧 Industrial / Medium-Large Enterprise

Est. VoLL: Est. medium-high VoLL

Diversified industry, logistics, processing -- significant continuity requirements

2,109 substations (24.8%) High/Critical: **342** (16.2%) Avg R: **0.383** Avg E: **0.810** / 0.10

Low 280 Med 1487 High 341 Crit 1

🏢 Commercial / SME-Dense

Est. VoLL: Est. medium VoLL

Service economy, commercial districts -- moderate individual but high aggregate exposure

2,131 substations (25.1%) High/Critical: **383** (18.0%) Avg R: **0.386** Avg E: **0.833** / 0.10

Low 286 Med 1462 High 381 Crit 2

🌾 Light / Agricultural / Remote

Est. VoLL: Est. low VoLL

Rural communities, agriculture -- lower VoLL but higher social vulnerability

2,114 substations (24.9%) High/Critical: **369** (17.5%) Avg R: **0.386** Avg E: **0.888** / 0.10

Low 279 Med 1466 High 368 Crit 1

Value of Lost Load (VoLL) context: AEMO's 2024 Value of Customer Reliability (VCR) report places average VoLL at A$33.46/kWh for commercial/industrial and A$19.37/kWh for residential customers across the NEM. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **0 substations** combine capital-intensive economic fabric (R3 ≥ 1.14) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **New South Wales (690), Queensland (469), Victoria (425)** — regions where industrial corridors depend on uninterrupted power for continuous processes (aluminium smelting, LNG liquefaction, mining beneficiation). A single hour of unplanned outage at these substations carries an estimated VoLL of A$20–40/kWh, compared to A$1–4/kWh in remote pastoral areas. The fleet-wide average energy poverty rate is 8.8%, but this masks sharp state-level variation: South Australia and Tasmania combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 State & SA4 Digital Readiness Ranking

SA4 regions ranked by average R7 modifier — lower values indicate weaker digital infrastructure (broadband penetration, smart meter coverage). Cross-referenced with average R_median to identify regions combining digital exposure with high grid stress.

WORST DIGITAL READINESS — TOP 15 SA4 REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Northern Territory ⚠ high R	1.0082
2	Western Australia ⚠ high R	1.0087
3	Victoria	1.0096
4	Tasmania	1.0098
5	South Australia ⚠ high R	1.0104
6	Queensland ⚠ high R	1.0106
7	New South Wales	1.0107
8	Australian Capital Territory	1.0115

SA4-level analysis reveals a clear digital divide mirroring Australia's metropolitan-regional connectivity gap. **Northern Territory** has the lowest average R7 (1.0082), while **Australian Capital Territory** leads at 1.0115 — a spread of 0.0033 across the R7 range. **2 SA4 regions** combine below-median digital readiness with above-median grid risk, making them priority targets for DNSP digitalisation investment and the Australian Government's Rewiring the Nation programme. The R7 modifier is intentionally narrow ([0.99, 1.05]) to reflect the current limitations of open broadband/smart meter data — as ACSC/SOCI Act compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

> **Inaugural edition — Baseline Snapshot (April 2026).** This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0101, median = 1.0100; 0 substations classified HIGH cyber-exposure; 711 blind-spot substations (High/Critical risk + below-median R7); 4 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging SA4 regions where broadband and smart meter rollouts (DNSP Advanced Metering Infrastructure programmes) translate into measurable R7 shifts. The next material R7 update is expected when ABS publishes updated Digital Inclusion Index data (anticipated Q3 2026).

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **25 verified public data sources** feeding 125 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

25

125 variables

WEEKLY / MONTHLY

5

High-frequency feeds

QUARTERLY / ANNUAL

18

Regular refresh cycle

STATIC / DERIVED

2

Standards + scenarios

Data Source Registry — May 2026

Loading source registry...

C.1 Update Frequency Timeline

Loading...

C.2 Fleet Score Snapshot

Loading...

Loading...

C.3 Changelog — v3.4 → v4.0.2

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SECTION D — MONTHLY DEEP-DIVE

The NSW–VIC Interconnector Corridor: Where Demand Surges Outrun Transmission Capacity

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** NSW–VIC Interconnector corridor · **002** South Australian DER Saturation · **003** Queensland Tropical Infrastructure Stress · **004** Metropolitan-Remote economic impact disparity · **005** Coal closure transition risk (Hunter Valley) · **006** Infrastructure age vs smart meter rollout · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** SA4 Grid Health Cards · **010** Academic validation and peer feedback · **011** Asia-Pacific replication feasibility · **012** Year in review.

D.1 Why the NSW–VIC Corridor, Why Now

NSW SUBSTATIONS	HIGH BAND	CORRIDOR MEDIAN R
2750	431	0.378

986 HV · 1764 MV

15.7% of corridor

Fleet median: 0.383

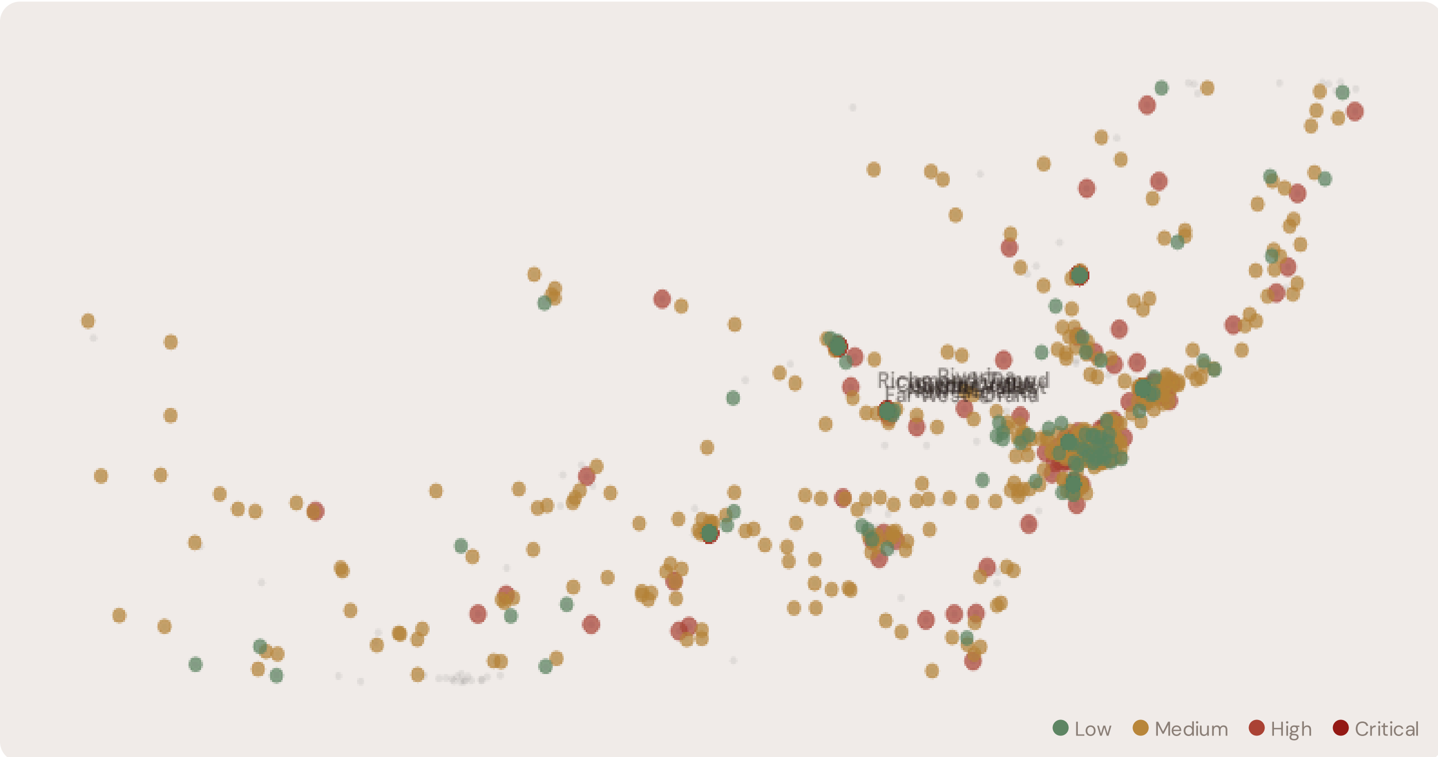
WORST SA4 REGION

Sydney-West

Avg R: 0.396 · 149 substations

The NSW corridor hosts **2750 substations** across multiple SA4 regions, forming the backbone of Australia's east-coast interconnected grid. Its risk profile is disproportionately concentrated in the upper bands: **431 substations (15.7%)** are classified High, with only **406** in the Low band. 48% of NSW substations score above the national fleet median of 0.383. The corridor median R of 0.378 reflects the tension between rapid coal-to-renewable transition, accelerating DER penetration, growing interconnector demand, and ageing infrastructure built for centralised baseload generation. The SSI flags continuity-of-supply deficits, infrastructure age mismatches, and the strain of bidirectional flows on a unidirectional network.

D.2 Corridor Map and Scoring



SUBSTATION	SA4 REGION	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Albury BSP 662	Sydney-SW	0.746	High	0.123	1.026	0.579	2.032	0.833	0.857	V, E, S, T
Dubbo ZSub 13	Capital Region	0.744	High	0.971	1.922	0.271	0.123	0.904	3.556	C, V, S, T
Bathurst BSP 681	Mid North Coast	0.716	High	0.611	2.165	0.521	1.241	0.281	4.508	C, V, E, T

SUBSTATION	SA4 REGION	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Tamworth BSP 725	Sydney-West	0.707	High	0.125	1.560	0.370	0.506	1.248	5.879	V, S, T
Wagga Wagga TStn 215	Illawarra	0.706	High	0.020	3.624	0.604	1.580	0.148	4.217	V, I, E, T
Newcastle TStn 180	Richmond-Tweed	0.700	High	0.243	0.314	0.901	0.696	1.218	3.517	I, E, S, T
Newcastle TStn 871	Sydney-East	0.699	High	0.666	0.895	0.736	2.063	0.158	5.572	C, V, I, E, T
Bathurst Sub 312	New England	0.696	High	0.431	2.532	0.105	0.392	0.777	6.258	V, S, T
Tamworth TStn 54	Illawarra	0.696	High	0.373	0.581	1.050	0.639	1.510	3.829	I, E, S, T
Tamworth Sub 5	Sydney-West	0.694	High	0.640	1.417	0.682	1.141	0.242	1.765	C, V, I, E, T

… 2740 additional substations — explore full corridor in Digital Twin →

D.3 Component Analysis

COMPONENT AVERAGES — NSW CORRIDOR VS FLEET

C — Continuity 0.266 vs 0.273 (-3%)

I — Infrastructure 0.321 vs 0.328 (-2%)

S — Saturation 0.400 vs 0.410 (-2%)

V — Voltage 0.782 vs 0.818 (-4%)

E — Economic 0.823 vs 0.824 (-0%)

T — Transition 1.615 vs 1.642 (-2%)

MODIFIER AVERAGES — NSW CORRIDOR VS FLEET

R3 Consequence	**0.948**	fleet 0.949 ↓
R4 Graph Crit.	**0.901**	fleet 0.900 ↓
R6a Restoration	**0.921**	fleet 0.921 ↓
R6b Climate Hazard	**NaN**	fleet NaN →
R7 Digital Read.	**1.011**	fleet 1.010 ↑

The dominant risk driver across the NSW corridor is **T (Transition)**, averaging 1.615 against a fleet average of 1.642 — occupying 3229% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.823 vs fleet 0.824 (823% of ceiling). Among modifiers, the R3 consequence multiplier averages 0.948 (fleet: 0.949), reflecting the economic significance of NSW's industrial base — from the Hunter Valley coal chain to Western Sydney logistics corridors. The R6b climate hazard modifier averages NaN, capturing the corridor's increasing exposure to extreme heat events and bushfire risk — a dimension invisible to every competing framework.

D.4 SA4 Region Breakdown

SA4 REGION RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.

Sydney-West 149 subs · avg R = 0.396 · H:31 M:96

Far West-Orana 161 subs · avg R = 0.395 · H:32 M:108

Hunter Valley 137 subs · avg R = 0.391 · H:24 M:100

Richmond-Tweed 169 subs · avg R = 0.386 · H:26 M:125

Sydney-SW 153 subs · avg R = 0.380 · H:25 M:109

New England 152 subs · avg R = 0.379 · H:23 M:110

Riverina 137 subs · avg R = 0.378 · H:20 M:97

Central Coast 161 subs · avg R = 0.378 · H:24 M:118

Newcastle 157 subs · avg R = 0.377 · H:21 M:114

Sydney-East 158 subs · avg R = 0.375 · H:25 M:105

Capital Region 158 subs · avg R = 0.373 · H:25 M:108

Sydney-City 150 subs · avg R = 0.370 · H:19 M:106

Sth Highlands 170 subs · avg R = 0.370 · H:27 M:118

Illawarra 154 subs · avg R = 0.367 · H:26 M:98

Murray 147 subs · avg R = 0.366 · H:21 M:104

Coffs Harbour 124 subs · avg R = 0.366 · H:23 M:75

Mid North Coast 159 subs · avg R = 0.365 · H:23 M:108

Sydney-North 154 subs · avg R = 0.361 · H:16 M:114

The SA4-level breakdown reveals significant intra-state variation. **Sydney-West** leads with a mean R of 0.396 across 149 substations, while **Sydney-North** scores 0.361 — a spread of 0.035 within a single state. This disparity underscores why state-level metrics (as used by AER performance reports) mask the true distribution of grid stress. The SSI's substation-level granularity reveals that NSW is not uniformly stressed but contains distinct corridors of concentrated risk — precisely the kind of spatial pattern that AEMO's Integrated System Plan and DNSP capital expenditure planning requires.

D.5 Implications

26 NSW substations score R ≥ 0.650, placing them in the upper quartile of national risk. For AEMO ISP investment prioritisation and Rewiring the Nation funding allocation, this analysis identifies the Sydney-West corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve — characterised by coal transition dependency, growing Western Sydney demand, and bushfire-prone supply corridors — are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

Asia-Pacific Context Card

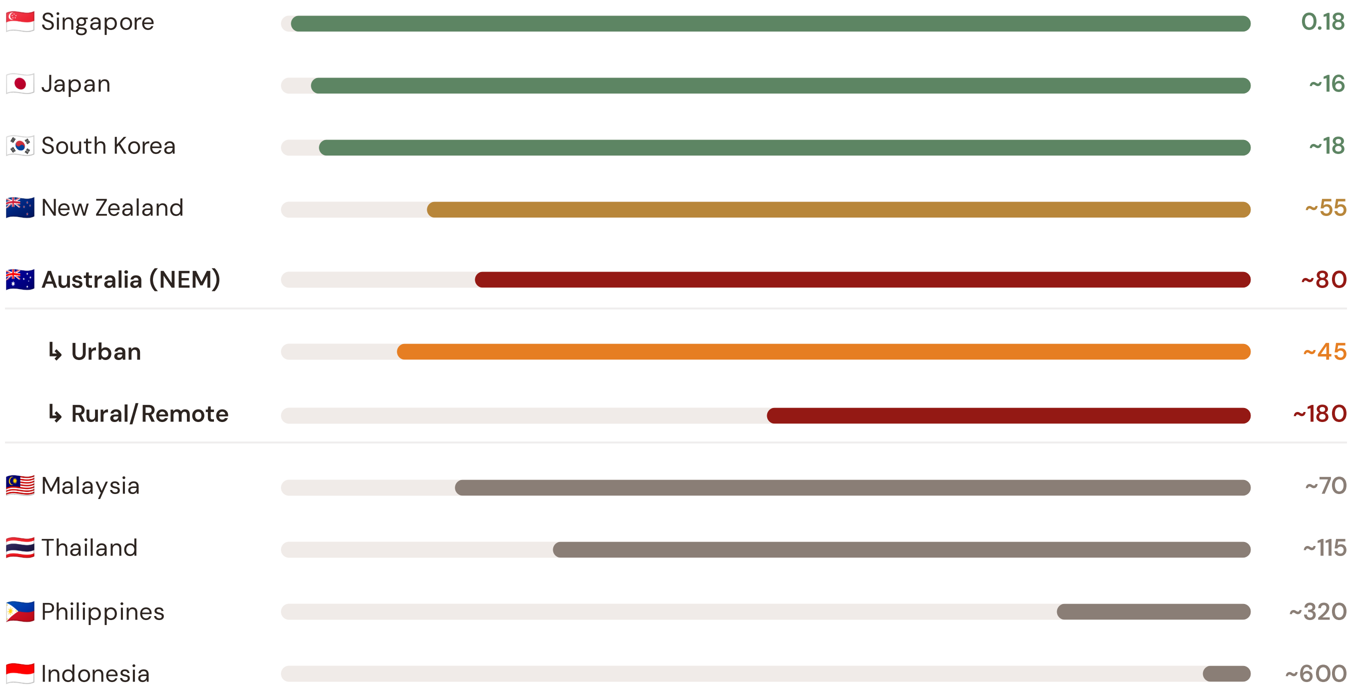
How this works: Each edition includes one Asia-Pacific Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the NSW–VIC corridor analysis hinges on Australia's continuity performance relative to regional peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Japan (003), CMIP6 climate hazard vs SE Asian peers (004), etc. The underlying data updates annually with AEMO/AER and national regulator publications.

ASIA-PACIFIC CONTEXT CARD — SAIDI BENCHMARKING

Ed. 001 · May 2026

Unplanned SAIDI (min/customer/year) — Selected Asia-Pacific Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



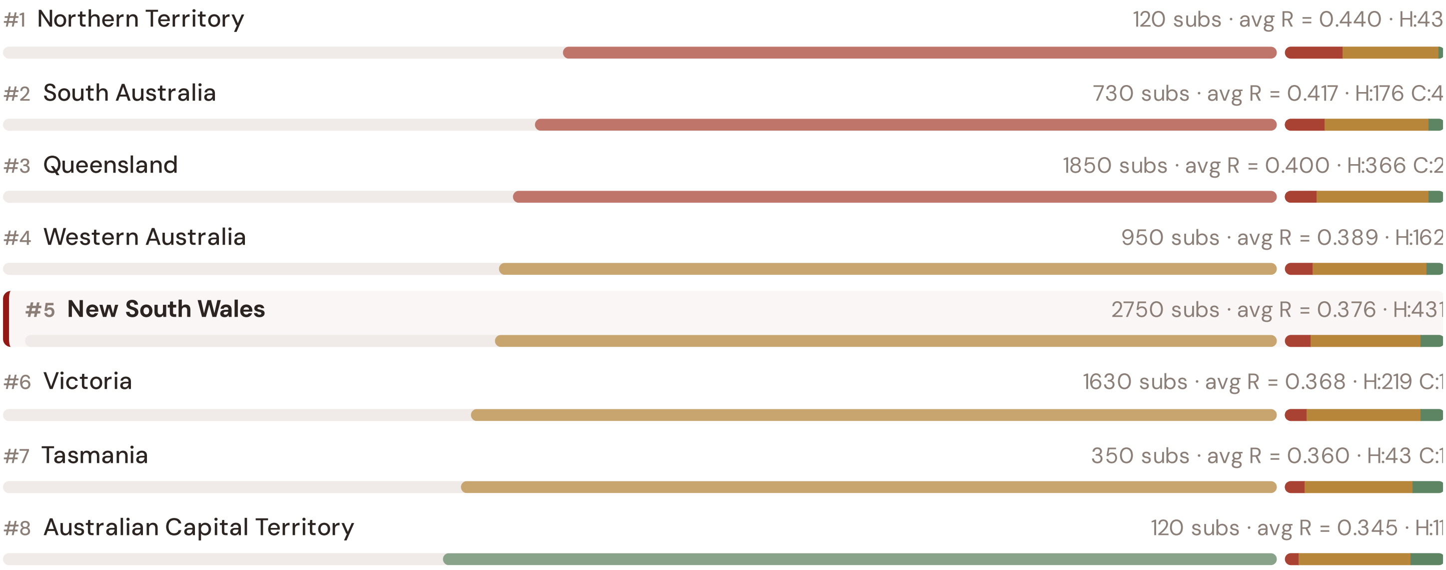
Source: AEMO Reliability Panel Annual Report (2024); AER State of the Energy Market (2025); World Bank ESMAP (2023); individual national regulator publications. Australia sub-categories from AER urban/rural/remote classifications. · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually.

This month's key insight: The NSW corridor deep-dive shows **1566 substations** (of 2750) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with Australia's rural/remote territory classification. The NSW corridor's average C of 0.266 is **1.0× the fleet average** of 0.273. In Asia-Pacific terms, these substations individually perform closer to Malaysian or Thai SAIDI levels (~70–115 min/customer/year) than to Australia's own NEM urban average (~45 min). The SSI reveals what aggregate AER statistics conceal: that Australia's mid-tier Asia-Pacific position is not a national condition but a statistical average of Singapore-quality urban performance and developing-nation-level remote performance — and the NSW corridor concentrates significant pockets of the latter.

State/Territory Risk Ranking — All 8 Australian States & Territories

Where does the NSW corridor sit within the national landscape? States sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



New South Wales ranks **#5 of 8 states/territories** by mean R_median. The three most stressed states are Northern Territory, South Australia, Queensland, while the three most resilient are Australian Capital Territory, Tasmania, Victoria. 4 of 8 states score above the fleet average of 0.385. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DNSP or national statistics — reveals the true spatial distribution of grid stress across Australia. It is precisely this substation-level granularity that positions the SSI as a tool for AEMO ISP / Rewiring the Nation / CER investment prioritisation: not treating entire states as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the DNSP/TNSP and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

Layer 1 — Data	Layer 2 — Valuation	Layer 3 — Neural Net	Layer 4 — Coverage	Layer 5 — Platform
SSI v4.0 component taxonomy, normalisation, MC engine	DNSP/TNSP + Community value stacks, TCO, real options	GBT+DNN ensemble, transfer learning, SHAP	Country prioritisation, DCI, registry, calibration	Products, API, revenue model, unit economics

LAYER 3 §0.5.5 · edition · May 2026

Uncertainty Quantification

Investment-grade predictions require calibrated confidence intervals, not just point estimates

A point estimate without uncertainty bounds is unusable for investment-grade analysis. The SSI-ENN produces calibrated P10/P50/P90 prediction intervals using two complementary approaches: quantile regression (the DNN head directly outputs multiple quantiles, trained with pinball loss) and conformal prediction (a distribution-free method that provides guaranteed coverage). The SSI's own Monte Carlo confidence intervals (CI_width, P5/P95) serve as the ground truth for calibration: the neural network's uncertainty estimates are validated against the analytical uncertainty from the 10,000-iteration copula engine.

KEY CONCEPTS

- → Quantile regression (pinball loss) + conformal prediction
- → Calibrated against SSI Monte Carlo ground truth
- → Per-country, per-voltage, per-band calibration verification
- → Investment-grade: P10/P50/P90 for all outputs

LIVE SSI DATA CONNECTION

The Australian training set comprises 8,500 substations with complete 95-feature vectors. Average CI_width of 0.190 provides the uncertainty ground truth for neural network calibration. The fleet's risk distribution — 1122 Low, 5919 Medium, 1451 High, 8 Critical — ensures balanced training across all risk bands.

Coming next month: LAYER 3 Explainability & SHAP Values — *Clients need to understand not just the prediction but WHY a substation ranks highly*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 003

Hunter Valley Coal Chain

Deep-dive into Hunter Valley's grid risk profile — substation map, component analysis, SA4 breakdown. Asia-Pacific Context Card: Eraring closure and REZ development interaction.

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Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent an AEMO participant, DNSP, TNSP, state government, research institution, or industry body and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 002

Published 14 May 2026 · SSI Index v4.0.2 · 8,500 substations · ~9,200 grid lines · 6 components · 20 metrics · 6 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5